

RAGHAV AWASTHI

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EDUCATION

University College London (UCL)

Bachelor of Science in Computer Science, First Class Honours

London, UK

Sep 2023 - Jun 2026

Dubai College

*A*A*A*A*A in A-Levels and 11 grade 9s in GCSEs*

Dubai, UAE

Aug 2016 - Jun 2023

EXPERIENCE

AI/ML Engineering Intern

Jul 2026 - Present

VotalAI

- Implemented feature-branch fixes for five security weakness classes across an AI gateway by hardening hCaptcha verification, backend URLs, subprocess model paths, tenant regexes and LLM tool-call validation against CWE-288, CWE-918, CWE-78, ReDoS and prompt injection.
- Implemented Votal, Nemotron and dual-model content-safety enforcement by integrating NVIDIA Nemotron 3.5 Content Safety into the model router and validating its live request/response contract.

Cybersecurity Analyst

Jun 2025 - Aug 2025

Barclays

Knutsford, UK

- Projected a 20-30% faster PQC migration across 30+ services by producing a 60-page PQC/HSM assessment, capability and gap matrix, and five-phase implementation roadmap.
- Cut certificate triage time 25% and flagged 30+ high-risk certificates by automating chain enrichment, root-cause classification, ServiceNow normalisation and risk scoring in Python.
- Increased high-signal detections 15-20% in pilot by identifying 100+ threat keywords and deploying the resulting detection logic to production.

Full-Stack Developer

Aug 2024 - Oct 2024

Adamas Data

London, UK

- Served 140M+ rows at median latency below 3 seconds by designing 30+ versioned FastAPI endpoints, indexing PostgreSQL/PostGIS queries and adding Dockerised CI/CD with rollback under 10 minutes.

AI/ML and Quantum Security Intern

Jun 2024 - Aug 2024

Tech Mahindra

Pune, India

- Demonstrated post-quantum key agreement by evaluating NIST finalist KEMs and prototyping hybrid TLS 1.3 with ML-KEM in Python.

PROJECTS

Quantum Proximity Gateway (IBM and UCL) | *Python, Rust, C++, Next.js*

Oct 2024 - May 2025

- Automated proximity-based PC authentication through a multi-factor edge system combining BLE presence and facial recognition with USB-HID emulation.

Post-Quantum Cryptography Research | *Pioneer Academics, C, Linux*

Feb 2022 - Jul 2022

- Produced a supervised research paper comparing code-, lattice- and multivariate-based NIST Round 3 PQC finalists through mathematical analysis and empirical C benchmarks.

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, Bash, Java, JavaScript/TypeScript, SQL

Security: PKI, HSMs, certificates, PQC, ML-KEM, TLS 1.3, application/API security, threat detection

Tools: OpenSSL, Linux, Docker, ServiceNow, Git, CI/CD, IBM Qiskit, Selenium

Engineering: FastAPI, REST APIs, microservices, PostgreSQL, Redis, testing

OPEN SOURCE AND LEADERSHIP

- Delivered merged contributions to Microsoft Terminal, mactop and Docusaurus; served as a UCL programming tutor, cricket vice-captain and technical writer.